

Part 1

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After training an MST dependency parser over NLTK data,
The results we got (when trying 2 seeds to validate):

Total sentences: 3914

Train set size: 3522

Test set size: 392

Feature vocabulary size: 1035910

	Seed A (42)	Seed B (123)
Iteration 1 loss	34/3522	37/3522
Iteration 2 loss	50/3522	46/3522
Test UAS	0.1813	0.1811

The results are still better than random guessing (which would be $1/\text{avg_of_words_per_Sentence}$, so $\sim 1/23=4.3\%$),
but the model overall performs poorly.

Part 2

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Results of the attention based parsing:

	UAS
Layer = 0	0.2304
Layer = 5	0.1377
Layer = 11	0.1201

Among the three attention-based settings, the best result was obtained with the first BERT layer, layer 0, which achieved $\text{UAS} = 0.2304$. This suggests that lower-layer attention captures more local and surface-level syntactic relations, which are often useful for dependency parsing. The middle and final layers performed worse, possibly because their attention patterns encode more abstract semantic information rather than direct head-dependent syntactic structure.

Results of both Parsers for comparison:

Model	UAS
Perceptron MST	0.1813
Attention (Best layer)	0.2304

The MST parser requires supervised dependency-parsing training data and a training phase, whereas the attention-based parser uses a pretrained BERT model directly and does not require additional parsing-specific training. In terms of inference cost, both parsers use Chu-Liu/Edmonds decoding, but the attention-based parser is more computationally expensive because it must run BERT on every sentence. The attention-based parser was more accurate in our experiment, suggesting that pretrained language-model attention contains some syntactic signal. This may happen because BERT is trained to predict words from context, and learning useful contextual representations often requires capturing relations between words that overlap with syntactic head-dependent structure.